

81. Let 3, 7, 11, 15, ..., 403 and 2, 5, 8, 11, ..., 404 be two arithmetic progressions. Then the sum, of the common terms in them, is equal to _____.

माना 3, 7, 11, 15, ..., 403 तथा 2, 5, 8, 11, ..., 404 दो समांतर श्रेढियाँ हैं। तो इनमें उभयनिष्ठ पदों का योग है _____

Ans. 6699

Sol. 3, 7, 11, 15, ..., 403

2, 5, 8, 11, ..., 404

So common term AP

11, 23, 35, ..., 395

$$\Rightarrow 395 = 11 + (n - 1)12$$

$$\Rightarrow n = 33$$

$$\text{Sum} = \frac{33}{2} [2 \times 11 + (32)12]$$

$$= \frac{33}{2} [22 + 384]$$

$$= 6699$$

82. The interior angles of a polygon with n sides, are in an A.P. with common difference 6° . If the largest interior angle of the polygon is 219° , then n is equal to _____.

n भुजाओं वाले एक बहुभुज के अंतः कोण 6° के सार्व अंतर की एक A.P. में हैं। यदि बहुभुज का सबसे बड़ा अंतःकोण 219° , तो n बराबर है _____

Ans. 20

Sol. n sides = n angles

$$d = 6$$

$$\text{Total sum of angles} = (n - 2)\pi$$

$$\left(\frac{n}{2}\right)(a + a + (n - 1)d) = (n - 2)\pi$$

$$\left(\frac{n}{2}\right)(a + 219^\circ) = (n - 2)(180^\circ)$$

$$\left(\frac{n}{2}\right)(219 - (n - 1)(6) + 219) = (n - 2)(180)$$

$$\Rightarrow (n) (438 - 6n + 6) = 360(n - 2)$$

$$\Rightarrow (n) (444 - 6n) = 360(n - 2)$$

$$\Rightarrow 444n - 6n^2 = 360n - 720$$

$$\Rightarrow 6n^2 - 84n - 720 = 0$$

$$\Rightarrow n^2 - 14n - 120 = 0$$

$$(n - 20)(n + 6) = 0$$

$$n = 20$$

83. Let $\alpha\beta\gamma = 45$; $\alpha, \beta, \gamma \in \mathbb{R}$. If $x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$ for some $x, y, z \in \mathbb{R}$, $x, y, z \neq 0$, then $6\alpha + 4\beta + \gamma$ is equal to _____.

माना $\alpha\beta\gamma = 45$; $\alpha, \beta, \gamma \in \mathbb{R}$ है। यदि कुछ $x, y, z \in \mathbb{R}$, $x, y, z \neq 0$, के लिए

$x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$ है, तो $6\alpha + 4\beta + \gamma$ बराबर है _____

Ans. 55

Sol. $\alpha\beta\gamma = 45$, $\alpha\beta\gamma \in \mathbb{R}$

$$x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$$

$$x, y, z \in \mathbb{R}, xyz \neq 0$$

$$\alpha x + y + 2z = 0$$

$$x + \beta y + 3z = 0$$

$$2x + 2y + \gamma z = 0$$

$$xyz \neq 0 \Rightarrow \text{non-trivial}$$

$$\begin{vmatrix} \alpha & 1 & 2 \\ 1 & \beta & 3 \\ 2 & 2 & \gamma \end{vmatrix} = 0$$

$$\Rightarrow \alpha(\beta\gamma - 6) - 1(\gamma - 6) + 2(2 - 2\beta) = 0$$

$$\Rightarrow \alpha\beta\gamma - 6\alpha - \gamma + 6 + 4 - 4\beta = 0$$

$$\Rightarrow 6\alpha + 4\beta + \gamma = 55$$

84. Let A be a 2×2 real matrix and I be the identity matrix of order 2. If the roots of the equation $|A - xI| = 0$ be -1 and 3 , then the sum of the diagonal elements of the matrix A^2 is _____.

माना A एक 2×2 का वास्तविक आव्यूह है तथा I , 2 कोटि का तत्समक आव्यूह है। यदि समीकरण $|A - xI| = 0$ के मूल -1 तथा 3 हैं, तो आव्यूह A^2 के विकर्ण के अवयवों का योग है _____

Ans. 10

Sol. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$|A - xI| = 0 \Rightarrow \begin{vmatrix} a-x & b \\ c & d-x \end{vmatrix} = 0$$

$$(a-x)(d-x) - bc = 0$$

$$x^2 - (a+d)x + ad - bc = 0$$

has roots -1 and 3

$$a + d = 2 \quad ad - bc = -3$$

$$A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix}$$

Sum of diagonal element of A^2

$$= a^2 + d^2 + 2bc$$

$$= (a+d)^2 + 2(bc - ad)$$

$$= 4 + 6 = 10$$

85. Let $a_1, a_2, \dots, a_{2024}$ be an Arithmetic Progression such that $a_1 + (a_5 + a_{10} + a_{15} + \dots + a_{2020}) + a_{2024} = 2233$. Then $a_1 + a_2 + a_3 + \dots + a_{2024}$ is equal to _____.

माना कि $a_1, a_2, \dots, a_{2024}$ एक समान्तर श्रेणी इस प्रकार है कि $a_1 + (a_5 + a_{10} + a_{15} + \dots + a_{2020}) + a_{2024} = 2233$ है। तब $a_1 + a_2 + a_3 + \dots + a_{2024}$ बराबर है _____.

Ans. 11132

Sol. $a_1 + a_5 + a_{10} + \dots + a_{2020} + a_{2024} = 2233$

In an A.P. the sum of terms equidistant from ends

is equal.

$$a_1 + a_{2024} = a_5 + a_{2020} = a_{10} + a_{2015} \dots$$

$$\Rightarrow 203 \text{ pairs}$$

$$\Rightarrow 203(a_1 + a_{2024}) = 2233$$

Hence,

$$S_{2024} = \frac{2024}{2}(a_1 + a_{2024})$$

$$= 1012 \times 11$$

$$= 11132$$

86. Let I be the identity matrix of order 3×3 and for the matrix $A = \begin{bmatrix} \lambda & 2 & 3 \\ 4 & 5 & 6 \\ 7 & -1 & 2 \end{bmatrix}$, $|A| = -1$. Let B be the inverse of the matrix $\text{adj}(A \text{adj}(A^2))$. Then $|(\lambda B + I)|$ is equal to _____.

माना I , कोटि 3×3 का तत्समक आव्यूह है और आव्यूह $A = \begin{bmatrix} \lambda & 2 & 3 \\ 4 & 5 & 6 \\ 7 & -1 & 2 \end{bmatrix}$ के लिए $|A| = -1$ है। माना आव्यूह B , आव्यूह $\text{adj}(A$

$\text{adj}(A^2))$ का व्युत्क्रम है। तो $|(\lambda B + I)|$ बराबर है _____.

Ans. 38

Sol. $|A| = \begin{vmatrix} \lambda & 2 & 3 \\ 4 & 5 & 6 \\ 7 & -1 & 2 \end{vmatrix} = -1$

$$\lambda(16) - 2(-34) + 3(-39) = -1$$

$$16\lambda = 48 \Rightarrow \lambda = 3$$

$$B^{-1} = \text{adj}(A \cdot \text{adj}(A^2))$$

$$\text{Let } C = A \cdot \text{adj}(A^2)$$

$$AC = A^2 \text{adj}(A^2) = |A|^2 \cdot I = I \Rightarrow C = A^{-1}$$

$$\text{Now } B^{-1} = \text{adj}(A^{-1}) = B = \text{adj}(A)$$

$$\text{Now } \lambda B + I \Rightarrow 3B + I$$

$$\text{Let } P = 3B + I$$

$$P = 3\text{adj}(A) + I$$

$$AP = 3A\text{adj}(A) + A$$

$$AP = 3|A| \cdot I + A$$

$$AP = A - 3I$$

$$|AP| = |A - 3I|$$

$$|A| \cdot |P| = \begin{vmatrix} 0 & 2 & 3 \\ 4 & 2 & 6 \\ 7 & -1 & -1 \end{vmatrix} = 38$$

$$|P| = -38$$

87. The roots of the quadratic equation $3x^2 - px + q = 0$ are 10^{th} and 11^{th} terms of an arithmetic progression with common difference $\frac{3}{2}$. If the sum of the first 11 terms of this arithmetic progression is 88, then $q - 2p$ is equal to _____.

द्विघात समीकरण $3x^2 - px + q = 0$ के मूल, सार्व अन्तर $\frac{3}{2}$ की एक समान्तर श्रेणी के $10^{\text{वें}}$ और $11^{\text{वें}}$ पद हैं। यदि इस समान्तर श्रेणी के पहले 11 पदों का योग 88 है, तो $q - 2p$ बराबर है _____.

Ans. 474

Sol. $S_{11} = \frac{11}{2}(2a + 10d) = 88$

$$a + 5d = 8$$

$$a = 8 - 5 \cdot \frac{3}{2}$$

$$= \frac{1}{2}$$

Roots are $T_{10} = a + 9d$

$$= \frac{1}{2} + 9 \cdot \frac{3}{2}$$

$$= 14$$

$$T_{11} = a + 10d = \frac{31}{2}$$

$$\frac{P}{3} = T_{10} + T_{11} = 14 + \frac{31}{2} = \frac{59}{2}$$

$$P = \frac{177}{2}$$

$$\& \frac{q}{3} = 7 \times 31$$

$$q = 651$$

Now,

$$q - 2p$$

$$\Rightarrow 651 - 177$$

$$\Rightarrow 474$$

88. The number of 3-digit numbers, that are divisible by 2 and 3, but not divisible by 4 and 9, is _____.

3-अंकों की संख्याओं जो कि 2 और 3 से भाज्य हैं परंतु 4 और 9 से भाज्य नहीं हैं, की संख्या है _____

Ans. 125

Sol. Required numbers are which are divisible by 6 are 102, 108, 114, 996

(Total numbers are 150)

Now

Numbers which are divisible by 36 are 108, 144, 972

(Total numbers are 25)

Therefore, the required three digit numbers which are divisible by 6 and not by 36 are $150 - 25 = 125$.

89. Let S_n be the sum to n -terms of an arithmetic progression 3, 7, 11, If $40 < \left(\frac{6}{n(n+1)} \sum_{k=1}^n S_k \right) < 42$, then n equals _____.

माना समान्तर श्रेणी 3, 7, 11, के प्रथम n पदों का योग S_n है। यदि $40 < \left(\frac{6}{n(n+1)} \sum_{k=1}^n S_k \right) < 42$ है, तो n बराबर है

Ans. 9

Sol. $S_k = 3 + 7 + 11 + \dots$ up to k terms

$$= \frac{k}{2} [6 + (k-1)4] = k(2k+1)$$

$$\sum_{k=1}^n (2k^2 + k) = \frac{2n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$

$$40 < 2(2n+1) + 3 < 42$$

$$35 < 4n < 37$$

$$8.75 < n < 9.25$$

$$n = 9$$

90. Let A be a 2×2 symmetric matrix such that $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ and the determinant of A be 1. If $A^{-1} = \alpha A + \beta I$, where I is an identity matrix of order 2×2 , then $\alpha + \beta$ equals _____.

यदि 2×2 कोटि का सममित आव्यूह A इस प्रकार है कि $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ तथा A की सारणिक 1 है। यदि $A^{-1} = \alpha A + \beta I$, जहाँ I कोटि

2×2 की तत्समक आव्यूह है, तो $\alpha + \beta$ बराबर है _____

Ans. 5

Sol. Let $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$

$$\text{Now } A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

$$a + b = 3, b + d = 7$$

$$ad - b^2 = 1$$

$$(3 - b)(7 - b) - b^2 = 1$$

$$21 - 10b = 1$$

$$\Rightarrow b = 2, a = 1, d = 5$$

$$\text{Now, } A^{-1} = \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha & 2\alpha \\ 2\alpha & 5\alpha \end{bmatrix} + \begin{bmatrix} \beta & 0 \\ 0 & \beta \end{bmatrix}$$

$$= \alpha + \beta = 5$$

91. Let $A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$. If the sum of the diagonal elements of A^{13} is 3^n , then n is equal to _____.

. माना $A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$ है। यदि A^{13} के मुख्य विकर्ण अवयवों का योग 3^n है, तब n बराबर होगा _____

Ans. 7

Sol. $A^2 = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ 3 & 0 \end{bmatrix} = 3 \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$

$$A^4 = 9 \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} = 9 \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$$

$$A^8 = 81 \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = 81 \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$$

$$A^{12} = 9 \times 81 \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} = 3^6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^{13} = 3^6 \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\text{Trace} = 3^6(2 + 1) = 3^7$$

92. If the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ has equal roots, where $a + c = 15$ and $b = \frac{36}{5}$, then $a^2 + c^2$ is equal to _____.

यदि समीकरण $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ के मूल बराबर हैं, जहाँ $a + c = 15$ और $b = \frac{36}{5}$ हैं, तो $a^2 + c^2$ बराबर है _____.

Ans. 117

Sol. sum of coefficient = $ab - ac + bc - ba + ca - cb$

$$= 0$$

So, one root is one

Product of root = 1

$$\frac{c(a - b)}{a(b - c)} = 1$$

$$ac - bc = ab - ac$$

$$2ac = b(a + c)$$

$$2ac = 15b \Rightarrow 2ac = 15 \times \frac{36}{5}$$

$$ac = 54$$

$$a^2 + c^2 = (a + c)^2 - 2ac$$

$$= 225 - 108 = 117$$

93. Let α, β be the roots of the equation $x^2 - ax - b = 0$ with $\text{Im}(\alpha) < \text{Im}(\beta)$. Let $P_n = \alpha^n - \beta^n$. If $P_3 = -5\sqrt{7}i, P_4 = -3\sqrt{7}i, P_5 = 11\sqrt{7}i$ and $P_6 = 45\sqrt{7}i$, then $|\alpha^4 + \beta^4|$ is equal to _____.

माना कि $\alpha, \beta, \text{Im}(\alpha) < \text{Im}(\beta)$ समीकरण $x^2 - ax - b = 0$ के मूल हैं। माना कि $P_n = \alpha^n - \beta^n$ है। यदि $P_3 = -5\sqrt{7}i, P_4 = -3\sqrt{7}i, P_5 = 11\sqrt{7}i$ और $P_6 = 45\sqrt{7}i$ हैं, तो $|\alpha^4 + \beta^4|$ बराबर है _____.

Ans. 31

Sol. $\alpha + \beta = a; \alpha\beta = -b$

$$P_6 = aP_5 + bP_4$$

$$45\sqrt{7}i = a \times 11\sqrt{7}i + b(-3\sqrt{7})i$$

$$11a - 3b = 45 \quad \dots(i)$$

$$\& P_5 = aP_4 + bP_3$$

$$11\sqrt{7}i = a(-3\sqrt{7})i + b(-5\sqrt{7}i)$$

$$11 = -3a - 5b \quad \dots(ii)$$

$$a = 3, b = -4$$

$$|\alpha^4 + \beta^4| = \sqrt{(\alpha^4 - \beta^4)^2 + 4(\alpha\beta)^4}$$

$$= \sqrt{961}$$

$$\Rightarrow 31$$

94. The number of singular matrices of order 2, whose elements are from the set $\{2, 3, 6, 9\}$, is _____.

कोटि 2 के अव्युत्क्रमणीय आव्यूहों, जिनके अवयव समुच्चय $\{2, 3, 6, 9\}$ से हैं, की संख्या है _____.

Ans. 36

Sol. $\begin{vmatrix} a & d \\ b & c \end{vmatrix} = ad - bc \Rightarrow ad = bc$

Case-I : Exactly 1 no. is used.

$$\Rightarrow \text{All singular} = {}^4C_1$$

Case-II : Exactly 2 no. is used

$$\Rightarrow {}^4C_2 \times 2 \times 2$$

Case-III : Exactly 3 no. is used

None will be singular

Case-IV : Exactly 4 no. is used

$$ad = bc$$

$$\Rightarrow 2 \times 9 = 3 \times 6$$

$$\begin{vmatrix} 9 & - \\ - & 2 \end{vmatrix} \Rightarrow {}^4C_1 \times 21$$

$$\text{Total} = 36$$

95. Let $\alpha, \beta \in \mathbb{N}$ be roots of the equation $x^2 - 70x + \lambda = 0$, where $\frac{\lambda}{2}, \frac{\lambda}{3} \notin \mathbb{N}$. If λ assumes the minimum possible

value, then $\frac{(\sqrt{\alpha-1} + \sqrt{\beta-1})(\lambda+35)}{|\alpha-\beta|}$ is equal to _____.

माना समीकरण $x^2 - 70x + \lambda = 0$, जहाँ $\frac{\lambda}{2}, \frac{\lambda}{3} \notin \mathbb{N}$ हैं, के मूल $\alpha, \beta \in \mathbb{N}$ हैं। λ के न्यूनतम संभव मानों के लिए,

$\frac{(\sqrt{\alpha-1} + \sqrt{\beta-1})(\lambda+35)}{|\alpha-\beta|}$ बराबर है _____।

Ans. 60

Sol. $x^2 - 70x + \lambda = 0$

$$\alpha + \beta = 70$$

$$\alpha\beta = \lambda$$

$$\therefore \alpha(70 - \alpha) = \lambda$$

Since, 2 and 3 does not divide λ

$$\therefore \alpha = 5, \beta = 65, \lambda = 325$$

By putting value of α, β, λ we get the required value of 60.

96. Let a, b, c be the lengths of three sides of a triangle satisfying the condition

$(a^2 + b^2)x^2 - 2b(a+c)x + (b^2 + c^2) = 0$. If the set of all possible values of x is the interval (α, β) , then $12(\alpha^2 + \beta^2)$ is equal to _____.

माना एक त्रिभुज की तीन भुजाओं की लम्बाइयाँ a, b, c हैं, जो $(a^2 + b^2)x^2 - 2b(a+c)x + (b^2 + c^2) = 0$ के संतुष्ट करती हैं।

यदि x के सभी संभव मानों का समुच्चय, अंतराल (α, β) है, तो $12(\alpha^2 + \beta^2)$ बराबर है _____

Ans. 36

Sol. $(a^2 + b^2)x^2 - 2b(a+c)x + (b^2 + c^2) = 0$

$$\Rightarrow (a^2x^2 - 2abx + b^2) + (b^2x^2 - 2bcx + c^2) = 0$$

$$\Rightarrow (ax - b)^2 + (bx - c)^2 = 0$$

$$\Rightarrow x = \frac{b}{a} \text{ and } x = \frac{c}{b} \Rightarrow b = ax$$

$$\Rightarrow \frac{b}{a} = \frac{c}{b} \Rightarrow b^2 = ac$$

Using triangular inequality

$$a + b > c, b + c > a, c + a > b$$

$$\Rightarrow a + ax > bx$$

$$\Rightarrow a + ax > ax^2 > 0$$

$$\Rightarrow x^2 - x - 1 < 0$$

Similarly, $ax + bx > a$

$$\Rightarrow x^2 + x - 1 > 0$$

$$c + a > b$$

$$\Rightarrow x^2 - x + 1 > 0 \text{ (always true)}$$

$$\Rightarrow x^2 - x - 1 < 0 \text{ and } x^2 + x - 1 > 0$$

$$\Rightarrow x \in \left(\frac{1-\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2} \right) \text{ and } x \in \left(\frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}+1}{2} \right)$$

$$\Rightarrow x \in \left(\frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}+1}{2} \right)$$

$$\Rightarrow \alpha^2 + \beta^2 = \left(\frac{\sqrt{5}-1}{2} \right)^2 + \left(\frac{\sqrt{5}+1}{2} \right)^2 = 3$$

$$\Rightarrow 12(\alpha^2 + \beta^2) = 36$$

$$\Rightarrow x^2 + x - 1 > 0$$

$$c + a > b$$

$$\Rightarrow x^2 - x + 1 > 0 \text{ (always true)}$$

$$\Rightarrow x^2 - x - 1 < 0 \text{ and } x^2 + x - 1 > 0$$

$$\Rightarrow x \in \left(\frac{1-\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2} \right) \text{ and } x \in \left(\frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}+1}{2} \right)$$

$$\Rightarrow x \in \left(\frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}+1}{2} \right)$$

$$\Rightarrow \alpha^2 + \beta^2 = \left(\frac{\sqrt{5}-1}{2} \right)^2 + \left(\frac{\sqrt{5}+1}{2} \right)^2 = 3$$

$$\Rightarrow 12(\alpha^2 + \beta^2) = 36$$

97. Let A be a 3×3 matrix of non-negative real elements such that $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Then the maximum value of $\det(A)$ is _____.

माना ऋणैत्तर वास्तविक अवयवों का 3×3 का एक आव्यूह A हैं तथा $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ है। तो $\det(A)$ का अधिकतम मान है

Ans. 27

Sol. Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Now

$$A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$$

$$a_{11} + a_{12} + a_{13} = 3$$

$$a_{21} + a_{22} + a_{23} = 3$$

$$a_{31} + a_{32} + a_{33} = 3$$

Now for maximum value of $\det(A) = a_{ij} \begin{cases} 0 & i \neq j \\ 3 & i = j \end{cases}$

$$\therefore |A| = 27$$

98. Let $A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, $B = [B_1, B_2, B_3]$, where B_1, B_2, B_3 are column matrices, and $AB_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $AB_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$,

$AB_3 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$. If $\alpha = |B|$ and β is the sum of all the diagonal elements of B , then $\alpha^3 + \beta^3$ is equal to _____.

माना $A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, $B = [B_1, B_2, B_3]$, जहाँ B_1, B_2, B_3 स्तम्भ आव्यूह हैं तथा $AB_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $AB_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$, $AB_3 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

हैं। यदि $\alpha = |B|$ तथा B के विकर्ण के सभी अवयवों का योग β है, तो $\alpha^3 + \beta^3$ बराबर है _____।

Ans. 28

Sol. Let $B_1 = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$

$$AB_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow A_1 = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 2p + r &= 1 \\ p + q &= 0 \text{ and } p + r = 0 \\ p &= 1, q = -1 \text{ and } r = -1 \end{aligned}$$

$$B_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\text{Let } B_2 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$AB_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 2x + z &= 2 \\ x + y &= 3 \text{ and } x + z = 0 \\ x &= 2, y = 1 \text{ and } z = -2 \end{aligned}$$

$$B_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

$$\text{Let } B_3 = \begin{bmatrix} \lambda \\ \mu \\ \phi \end{bmatrix}$$

$$AB_3 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda \\ \mu \\ \phi \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 2\lambda + \phi &= 3 \\ \lambda + \mu &= 2 \\ \lambda + \phi &= 1 \\ \lambda &= 2, \mu = 0 \text{ and } \phi = -1 \end{aligned}$$

$$B_3 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \Rightarrow B = \begin{bmatrix} 1 & 2 & 2 \\ -1 & 1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$$\alpha = |B| = 3$$

$$\beta = 1 + 1 - 1 = 1$$

$$\alpha^3 + \beta^3 = (3)^3 + 1^3 = 28$$

99. Let $A = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$. If for some $\theta \in (0, \pi)$, $A^2 = A^T$, then the sum of the diagonal elements of the matrix $(A + I)^3 + (A - I)^3 - 6A$ is equal to _____.

माना $A = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$ है। यदि किसी $\theta \in (0, \pi)$ के लिए $A^2 = A^T$ है, तो आव्यूह $(A + I)^3 + (A - I)^3 - 6A$ के विकर्ण के अवयवों का योग है _____.

Ans. 6

Sol. $A = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$

$$\theta \in (0, \pi)$$

$$A^2 = A^T$$

$$\text{Trace } \{(A + I)^3 + (A - I)^3 - 6A\} = ?$$

Sol by observation A is orthogonal

$$\therefore A^T = A^{-1}$$

$$\Rightarrow A^2 = A^{-1}$$

$$\Rightarrow A^3 = I$$

$$\therefore B = (A + I)^3 + (A - I)^3 - 6A$$

$$= 2(A^3 + 3A) - 6A$$

$$= 2A^3$$

$$= 2I$$

$$\therefore \text{Trace of } B = 2 + 2 + 2$$

$$= 6$$

100. Let $S = \{m \in \mathbb{Z} : A^{m^2} + A^m = 3I - A^{-6}\}$, where $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$. Then $n(S)$ is equal to _____.

माना कि $S = \{m \in \mathbb{Z} : A^{m^2} + A^m = 3I - A^{-6}\}$ है, जहाँ $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$ है। तब $n(S)$ बराबर है _____.

Ans. 2

Sol. $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$

$$\text{adj}(A) = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

$$|A| = 1$$

$$A^{-1} = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

$$(A^{-1})^2 = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix}$$

$$(A^{-1})^4 = \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 4 \\ -4 & 5 \end{bmatrix}$$

$$(A^{-1})^6 = \begin{bmatrix} -3 & 4 \\ -4 & 5 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & 6 \\ -6 & 7 \end{bmatrix}$$

$$A^{m^2} + A^m = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} -5 & 6 \\ -6 & 7 \end{bmatrix}$$

$$A^{m^2} + A^m = \begin{bmatrix} 8 & -6 \\ 6 & -4 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -4 \\ 4 & -3 \end{bmatrix}$$

$$m = 2$$

$$A^m = \begin{bmatrix} m+1 & -m \\ m & -(m-1) \end{bmatrix}$$

$$A^{m^2} + A^m = \begin{bmatrix} 8 & -6 \\ 6 & -4 \end{bmatrix}$$

$$m^2 + 1 + m + 1 = 8$$

$$m^2 + m - 6 = 0$$

$$(m+3)(m-2) = 0 \Rightarrow m = -3, m = 2$$